

11.6 Homework: Mixed Review — Markscheme

La Scuola d'Italia / Dr. Huson / IB Math

7 March 2026

Section A

1. (a) use of GDC to give **(M1)**

$$r = 0.883529...$$

$$r = 0.884$$

A1

Note: Award the **(M1)** for any correct value of r , a , b or $r^2 = 0.780624...$ seen in part (a) or part (b).

[2 marks]

- (b) $a = 1.36609...$, $b = 64.5171...$

$$a = 1.37, b = 64.5$$

A1

[1 mark]

- (c) attempt to find their difference **(M1)**

$$5 \times 1.36609... \text{ OR } 1.36609...(h+5) + 64.5171... - (1.36609...h + 64.5171...)$$

$$6.83045...$$

$$= 6.83 \text{ (6.85 from 1.37)}$$

the student could have expected her score to increase by 7 marks.

A1

Note: Accept an increase of 6, 6.83 or 6.85.

[2 marks]

Total [5 marks]

Section A

1. (a) EITHER

uses the cosine rule

(M1)

$$AB^2 = 5^2 + 5^2 - 2 \times 5 \times 5 \times \cos 1.9$$

(A1)

OR

uses right-angled trigonometry

(M1)

$$\frac{AB}{2}$$

$$= \sin 0.95$$

(A1)

OR

uses the sine rule

(M1)

$$\alpha = \frac{1}{2}(\pi - 1.9) (= 0.6207\dots)$$

$$\frac{AB}{\sin 1.9} = \frac{5}{\sin 0.6207\dots}$$

(A1)

THEN

$$AB = 8.13415\dots$$

$$AB = 8.13 \text{ (m)}$$

A1

[3 marks]

continued...

Question 1 continued

- (b) let the shaded area be A

METHOD 1

attempt at finding reflex angle

(M1)

$$\hat{AOB} = 2\pi - 1.9 \quad (= 4.3831\dots)$$

substitution into area formula

(A1)

$$A = \frac{1}{2} \times 5^2 \times 4.3831\dots \quad \text{OR} \quad \left(\frac{2\pi - 1.9}{2\pi} \right) \times \pi (5^2)$$

$$= 54.7898\dots$$

$$= 54.8 \text{ (m}^2\text{)}$$

A1

METHOD 2

let the area of the circle be A_C and the area of the unshaded sector be A_U

$$A = A_C - A_U$$

(M1)

$$A = \pi \times 5^2 - \frac{1}{2} \times 5^2 \times 1.9 \quad (= 78.5398\dots - 23.75)$$

(A1)

$$= 54.7898\dots$$

$$= 54.8 \text{ (m}^2\text{)}$$

A1

[3 marks]

Total [6 marks]

2. METHOD 1

recognises that $g(x) = \int (3x^2 + 5e^x) dx$ **(M1)**

$$g(x) = x^3 + 5e^x (+C) \quad \text{A1(A1)}$$

Note: Award **A1** for each integrated term.

substitutes $x = 0$ and $y = 4$ into their integrated function (must involve $+C$) **(M1)**

$$4 = 0 + 5 + C \Rightarrow C = -1$$

$$g(x) = x^3 + 5e^x - 1 \quad \text{A1}$$

METHOD 2

attempts to write both sides in the form of a definite integral **(M1)**

$$\int_0^x g'(t) dt = \int_0^x (3t^2 + 5e^t) dt \quad \text{(A1)}$$

$$g(x) - 4 = x^3 + 5e^x - 5e^0 \quad \text{(A1)(A1)}$$

Note: Award **A1** for $g(x) - 4$ and **A1** for $x^3 + 5e^x - 5e^0$.

$$g(x) = x^3 + 5e^x - 1 \quad \text{A1}$$

[5 marks]

3. (a) attempt to use $u_1 + (n-1)d = 0$ (M1)

$$60 - 2.5(k-1) = 0$$

$$k = 25$$

A1

[2 marks]

- (b) **METHOD 1**

attempting to express S_n in terms of n (M1)

use of a graph or a table to attempt to find the maximum sum (M1)

$$= 750 \quad \text{A1}$$

METHOD 2

EITHER

recognizing maximum occurs at $n = 25$ (M1)

$$S_{25} = \frac{25}{2}(60 + 0), \quad S_{25} = \frac{25}{2}(2 \times 60 + 24 \times -2.5) \quad \text{(A1)}$$

OR

attempting to calculate S_{24} (M1)

$$S_{24} = \frac{24}{2}(2 \times 60 + 23 \times -2.5) \quad \text{(A1)}$$

THEN

$$= 750 \quad \text{A1}$$

[3 marks]

Total [5 marks]

2. (a) 9% (accept 0.09)

A1

[1 mark]

(b) $t = 5$ (seen anywhere)

(A1)

24961.28...

25000 (dollars)

A1

[2 marks]

continued...

Question 2 continued

(c) **EITHER**

$$n = 5$$

$$I\% = 3$$

$$PV = (\mp)15000$$

$$P/Y = 1$$

$$C/Y = 1$$

(A1)

Note: Award **(A1)** for use of a financial app in their technology with all entries correct.

$$(\Rightarrow FV = (\pm)17389.11\dots)$$

OR

$$15000\left(1 + \frac{3}{100}\right)^5 (=17389.11\dots)$$

(A1)

THEN

subtracting their value from their answer to part (b)

(M1)

7572.17 ...

7570 (dollars)

A1

[3 marks]

Total [6 marks]

3.

Note: In this question, do not penalise incorrect use of strict inequality signs.

Let X = mass of a bag of sugar

(a) evidence of identifying the correct area **(M1)**

$$P(X < 995) = 0.0765637...$$

$$= 0.0766$$

A1

[2 marks]

(b) 0.0766×100
 ≈ 8

A1

[1 mark]

Note: Accept 7.66.

(c) recognition that $P(X > 1005 | X \geq 995)$ is required **(M1)**

$$\frac{P(X \geq 995 \cap X > 1005)}{P(X \geq 995)}$$

$$\frac{P(X > 1005)}{P(X \geq 995)}$$

(A1)

$$\frac{0.0765637...}{1 - 0.0765637...} \left(= \frac{0.0765637...}{0.923436...} \right)$$

$$= 0.0829$$

A1

[3 marks]

Total [6 marks]

4. (a) recognize that the variable has a Binomial distribution (M1)

$$X \sim B(30, 0.05)$$

attempt to find $P(X \geq 1)$ (M1)

$$1 - P(X = 0) \text{ OR } 1 - 0.95^{30} \text{ OR } 1 - 0.214638... \text{ OR } 0.785361...$$

Note: The two **M** marks are independent of each other.

$$P(X \geq 1) = 0.785 \quad \text{A1}$$

[3 marks]

- (b) recognition of conditional probability (M1)

$$P(X \leq 2 | X \geq 1) \text{ OR } P(\text{at most 2 defective} | \text{at least 1 defective})$$

Note: Recognition must be shown in context either in words or symbols but not just $P(A | B)$.

$$\frac{P(1 \leq X \leq 2)}{P(X \geq 1)} \text{ OR } \frac{P(X = 1) + P(X = 2)}{P(X \geq 1)} \quad \text{(A1)}$$

$$\frac{0.597540...}{0.785361...} \text{ OR } \frac{0.812178... - 0.214638...}{0.785361...} \text{ OR } \frac{0.338903... + 0.258636...}{0.785361...} \quad \text{(A1)}$$

$$= 0.760847...$$

$$P(X \leq 2 | X \geq 1) = 0.761 \quad \text{A1}$$

[4 marks]

Total [7 marks]

5. recognition that initial population is 15000 (seen anywhere) **(A1)**

$$P(0) = 15000 \text{ OR } 0.11 \times 15000 \text{ OR } 0.89 \times 15000$$

population after 11% decrease is $15000 \times 0.89 (=13350)$ **(A1)**

recognizing that $t = 8$ on 1 January 2022 (seen anywhere) **(A1)**

substitution of their value of t for 1 January 2022 and their value of $P(8)$ into the model **(M1)**

$$15000 \times 0.89 = 15000e^{8k} \text{ OR } 13350 = 15000e^{8k}$$

$$k = \frac{\ln 0.89}{8} (-0.014566) \text{ **(A1)**}$$

substitution of $t = 2041 - 2014 (= 27)$ and their value for k into the model **(M1)**

$$P(27) = 15000e^{-0.0145... \times 27}$$

10122.3...

$$P(27) = 10100 \text{ (10122)} \text{ **A1**}$$

Total [7 marks]

5. (a) attempting to find the vertex (M1)

$$x=1 \text{ OR } y=-5 \text{ OR } f(x)=6(x-1)^2-5$$

range is $y \geq -5$

A1

[2 marks]

- (b) **METHOD 1**

$$(g \circ f)(x) = -(6x^2 - 12x + 1) + c \quad (= -(6(x-1)^2 - 5) + c) \quad (A1)$$

EITHER

relating to the range of f OR attempting to find $g(-5)$ (M1)

$$5 + c \leq 0 \quad (A1)$$

OR

attempting to find the discriminant of $(g \circ f)(x)$ (M1)

$$144 + 24(c-1) \leq 0 \quad (120 + 24c \leq 0) \quad (A1)$$

THEN

$$c \leq -5 \quad A1$$

[4 marks]

METHOD 2

vertical reflection followed by vertical shift (M1)

new vertex is $(1, 5+c)$ (A1)

$$5 + c \leq 0 \quad (A1)$$

$$c \leq -5 \quad A1$$

[4 marks]

Total [6 marks]

5. (a) recognizing at rest $v = 0$ (M1)
 $t = 3.34692...$
 $t = 3.35$ (seconds) A1

Note: Award (M1)A0 for additional solutions to $v = 0$ eg $t = -0.205$ or $t = 6.08$.

[2 marks]

- (b) recognizing particle changes direction when $v = 0$ OR when $t = 3.34692...$ (M1)
 $a = -4.71439...$
 $a = -4.71(\text{ms}^{-2})$ A2

[3 marks]

- (c) distance travelled = $\int_0^6 |v| dt$ OR
 $\int_0^{3.34...} (e^{\sin(t)} + 4\sin(t)) dt - \int_{3.34...}^6 (e^{\sin(t)} + 4\sin(t)) dt$ (=14.3104...+6.44300...) (A1)
 $= 20.7534...$
 $= 20.8$ (metres) A1

[2 marks]

Total [7 marks]

4. (a) evidence of attempting to find correct area under normal curve (M1)

$$P(W > 210) \text{ OR sketch}$$

$$P(W > 210) = 0.115069...$$

$$P(W > 210) = 0.115$$

A1

[2 marks]

- (b) recognizing $P(W < w) = 1 - P(w < W < 210) - P(W > 210)$ (M1)

$$P(W < w) = 1 - 0.8 - 0.115069...$$

$$P(W < w) = 0.084930...$$

$$P(W < w) = 0.0849$$

A1

[2 marks]

- (c) evidence of attempting to use inverse normal function (M1)

$$w = 197.136...$$

$$w = 197 \text{ (grams)}$$

A1

[2 marks]

- (d) recognition of binomial distribution (M1)

$$X \sim B(10, 0.0849302...)$$

$$P(X = 1) = 0.382076...$$

$$P(X = 1) = 0.382$$

A1

[2 marks]

Total [8 marks]

5. (a) attempt to use the binomial expansion of $(x+h)^8$ (M1)

$${}^8C_0x^8h^0 + {}^8C_1x^7h^1 + {}^8C_2x^6h^2 + \dots$$

(i) $a = 8h$ (accept ${}^8C_1 h$) A1

(ii) $b = 28h^2$ (accept ${}^8C_2 h^2$) A1

(iii) $d = 70h^4$ (accept ${}^8C_4 h^4$) A1

[4 marks]

- (b) recognition that there is a common ratio between their terms (M1)

$$8h \times r = 28h^2 \text{ OR } 28h^2 \times r = 70h^4 \text{ OR } 8h \times r^2 = 70h^4$$

correct equation in terms of h (A1)

$$\frac{28h^2}{8h} = \frac{70h^4}{28h^2} \text{ (or equivalent)}$$

$h = 1.4$ A1

[3 marks]

Total [7 marks]

8. (a) (i) 32 (cm) **A1**

(ii) $h_A(0) = \sin(6) + 27$ **(M1)**

$$= 26.7205\dots$$

$= 26.7$ (cm) **A1**

[3 marks]

(b) attempts to solve $h_A(t) = h_B(t)$ for t **(M1)**

$$t = 4.00746\dots, 4.70343\dots, 5.88332\dots$$

$t = 4.01, 4.70, 5.88$ (weeks) **A2**

[3 marks]

continued...

Question 8 continued

(c) recognises that $h_A'(t)$ and $h_B'(t)$ are required (M1)

attempts to solve $h_A'(t) = h_B'(t)$ for t (M1)

$t = 1.18879\dots$ and $2.23598\dots$ OR $4.33038\dots$ and $5.37758\dots$ OR $7.47197\dots$ and $8.51917\dots$ (A1)

Note: Award full marks for $t = \frac{4\pi}{3} - 3, \frac{5\pi}{3} - 3, \left(\frac{7\pi}{3} - 3, \frac{8\pi}{3} - 3, \frac{10\pi}{3} - 3, \frac{11\pi}{3} - 3\right)$.

Award subsequent marks for correct use of these exact values.

$1.18879\dots < t < 2.23598\dots$ OR $4.33038\dots < t < 5.37758\dots$ OR
 $7.47197\dots < t < 8.51917\dots$

(A1)

attempts to calculate the total amount of time

(M1)

$$3(2.2359\dots - 1.1887\dots) \left(= 3 \left(\left(\frac{5\pi}{3} - 3 \right) - \left(\frac{4\pi}{3} - 3 \right) \right) \right)$$

$$= 3.14 (= \pi) \text{ (weeks)}$$

A1

[6 marks]

Total [12 marks]